The Maximum Likelihood Estimation For The Variance Is Biased

BIO210 Biostatistics

Extra Reading Material for Lecture 18

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The random variable $\boldsymbol{X}$ denotes certain metric (*e.g.* height, weight) we are interested in from a population, and $\boldsymbol{X} \sim \mathcal{N}(\mu, \sigma^2)$. We draw a random sample of size n from the population. Like we discussed during the lecture, a random sample of size n can be thought as n **i.i.d.** random variables. That is:

$$\boldsymbol{X}_1, \boldsymbol{X}_2, \boldsymbol{X}_3, \cdots, \boldsymbol{X}_n \sim \mathcal{N}(\mu, \sigma^2)$$

We have seen that the maximum likelihood estimator for σ^2 is:

$$\hat{\boldsymbol{\sigma}^2} = \frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{X}_i - \bar{\boldsymbol{X}})^2$$

Then, what is $\mathbb{E}\left[\hat{\boldsymbol{\sigma}^2}\right]$? If $\mathbb{E}\left[\hat{\boldsymbol{\sigma}^2}\right] = \sigma^2$, it is an unbiased estimator. Otherwise, it is a biased one.

Now let's have a look.

$$\mathbb{E}\left[\hat{\boldsymbol{\sigma}^2}\right] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{X}_i - \bar{\boldsymbol{X}})^2\right] = \frac{1}{n}\mathbb{E}\left[\sum_{i=1}^{n}(\boldsymbol{X}_i^2 - 2\bar{\boldsymbol{X}}\boldsymbol{X}_i + \bar{\boldsymbol{X}}^2)\right]$$

$$= \frac{1}{n}\mathbb{E}\left[\sum_{i=1}^{n}\boldsymbol{X}_i^2 - 2\bar{\boldsymbol{X}}\sum_{i=1}^{n}\boldsymbol{X}_i + \sum_{i=1}^{n}\bar{\boldsymbol{X}}^2\right]$$

Note that: $\sum_{i=1}^{n}\boldsymbol{X}_i = n\bar{\boldsymbol{X}}$. Since $\bar{\boldsymbol{X}}$ remains the same for each i, we have $\sum_{i=1}^{n}\bar{\boldsymbol{X}}^2 = n\bar{\boldsymbol{X}}^2$. Replacing the blue terms above, we have:

$$\mathbb{E}\left[\hat{\boldsymbol{\sigma}^2}\right] = \frac{1}{n}\mathbb{E}\left[\sum_{i=1}^{n}\boldsymbol{X}_i^2 - 2\bar{\boldsymbol{X}}\cdot n\bar{\boldsymbol{X}} + n\bar{\boldsymbol{X}}^2\right] = \frac{1}{n}\mathbb{E}\left[\sum_{i=1}^{n}\boldsymbol{X}_i^2 - n\bar{\boldsymbol{X}}^2\right]$$

$$= \frac{1}{n}\left(\mathbb{E}\left[\sum_{i=1}^{n}\boldsymbol{X}_i^2\right] - \mathbb{E}\left[n\bar{\boldsymbol{X}}^2\right]\right) \tag{1}$$

Since $\mathbb{V}\mathrm{ar}\left(\boldsymbol{X}\right) = \mathbb{E}\left[\boldsymbol{X}^2\right] - \left(\mathbb{E}\left[\boldsymbol{X}\right]\right)^2$, so we have $\mathbb{E}\left[\boldsymbol{X}^2\right] = \mathbb{V}\mathrm{ar}\left(\boldsymbol{X}\right) + \left(\mathbb{E}\left[\boldsymbol{X}\right]\right)^2$,

then,

$$\begin{aligned}\mathbb{E}\left[\sum_{i=1}^{n}\boldsymbol{X}_i^2\right] &= \mathbb{E}\left[\boldsymbol{X}_1^2\right]+\mathbb{E}\left[\boldsymbol{X}_2^2\right]+\mathbb{E}\left[\boldsymbol{X}_3^2\right]+\cdots+\mathbb{E}\left[\boldsymbol{X}_n^2\right]\\ &= \mathbb{V}\mathrm{ar}\left(\boldsymbol{X}_1\right)+\left(\mathbb{E}\left[\boldsymbol{X}_1\right]\right)^2+\mathbb{V}\mathrm{ar}\left(\boldsymbol{X}_2\right)+\left(\mathbb{E}\left[\boldsymbol{X}_2\right]\right)^2+\cdots\\ &\quad+\mathbb{V}\mathrm{ar}\left(\boldsymbol{X}_n\right)+\left(\mathbb{E}\left[\boldsymbol{X}_n\right]\right)^2\\ &= \sigma^2+\mu^2+\sigma^2+\mu^2+\cdots+\sigma^2+\mu^2\\ &= n\sigma^2+n\mu^2 \end{aligned} \tag{2}$$

Putting equation (2) into equation (1), we have:

$$\begin{aligned}\mathbb{E}\left[\hat{\boldsymbol{\sigma}^2}\right] &= \sigma^2+\mu^2-\frac{1}{n}\cdot\mathbb{E}\left[n\bar{\boldsymbol{X}}^2\right]=\sigma^2+\mu^2-\mathbb{E}\left[\bar{\boldsymbol{X}}^2\right]\\ &= \sigma^2+\mu^2-(\sigma_{\bar{X}}^2+\mu_{\bar{X}}^2)\end{aligned} \tag{3}$$

According to the central limit theorem, we have $\mu_{\bar{\boldsymbol{X}}}=\mu$ and $\sigma_{\bar{\boldsymbol{X}}}^2=\dfrac{\sigma^2}{n}$. Therefore, equation (3) becomes:

$$\mathbb{E}\left[\hat{\boldsymbol{\sigma}^2}\right]=\sigma^2+\mu^2-\frac{\sigma^2}{n}-\mu^2=\frac{n-1}{n}\sigma^2\neq\sigma^2$$

Hence, it is not an unbiased estimator.